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BE (4th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical, EEE

ELECTRICAL MEASUREMENTS & MEASURING INSTRUMENTS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d).

Q. 1. (a) Define bridge sensitivity. 2

(b) Draw the circuit of Kelvin's Double bridge used for measurement of low resistance. Derive formula for unknown resistance. 7

(c) With neat labelled diagram describe the construction and working of Megger. 7

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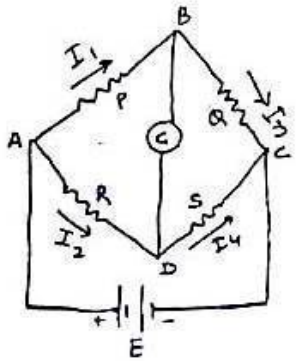
(d) In a wheat stone bridge the values of bridge elements are P = 100 Ω, Q = 1,000 Ω, S = 2,000 Ω and R = 202 Ω. The battery voltage is 5V and its internal resistance is negligible. The galvanometer used is of sensitivity 5 mm/μA and an internal resistance of 200 Ω. Determine the deflection of galvanometer caused by the 2 Ω unbalance in arm AD. Also determine the sensitivity of the bridge in terms of deflection per unit change in resistance. 7



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1.2. (a) Draw the circuit of that bridge which is useful in measuring high - Q coils. 2

(b) Describe the bridge used for the measurement of frequency with the derivation of balance equation. 7

(c) Explain Heaviside mutual inductance bridge for determination of mutual inductance of a coil. How self inductance of coil can be determined with the help of mutual inductance. 7

(d) Explain the function and working of wagner earth device. 7

2.3. (a) Define mechanical damping. 2

(b) Explain Wattmeter method for measurement of flux using Lloyd Fisher square. 7

(c) Explain the construction and working of a D'Arsonval galvanometer. 7

(d) Explain the ac potentiometer method for determination of iron losses. 7

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Q. 4. (a) Why damping is required.

(b) Explain with the help of diagram t construction and working of single pha electrodynamometer power factor meter.

(c) Derive the general torque equation for mov iron instruments. List their advantages.

(d) Describe the construction and working rotating and static type phase sequen indicators.

Q. 5. (a) How creeping can be avoided in ene meter.

(b) With the help of neat diagram explain Electrodynamometer wattmeter.

(c) Explain the method for measurement of pow by two wattmeter method in 3-ϕ circ Illustrate with the help of phasor diagram fo balanced star connected load.

(d) Explain the methods of testing of ene meters.

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